Symmetry: Even, Odd, or Neither

TEACHER KEY

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Even function: a function symmetric about the y-axis; algebraically, $f(-x) = f(x)$ for every x.

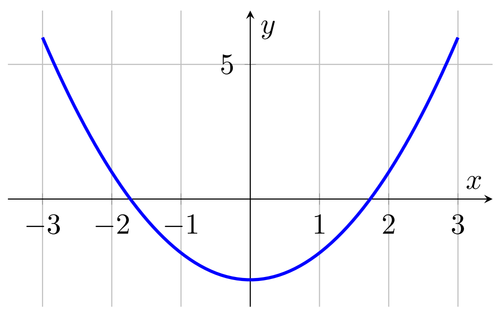
Odd function: a function symmetric about the origin; algebraically, $f(-x) = -f(x)$ for every x.

We do together. Determine whether $f(x) = x^2 - 3$ is even, odd, or neither.

Step 1. Substitute $-x$ for x: $f(-x) = (-x)^2 - 3 =$ {{$x^2 - 3$—2.4cm}}.

Step 2. Compare this result to the original $f(x) = x^2 - 3$: they are equal.

Step 3. Since $f(-x) = f(x)$, the function is even.



$y = x^2 - 3$ — the y-axis is a mirror line, confirming it's even.

You Try. Solve each. Show every step, then name the answer.

a) Determine whether $f(x) = x^3$ is even, odd, or neither. odd: $f(-x) = (-x)^3 = -x^3 = -f(x)$

b) Determine whether $f(x) = x^2 + x$ is even, odd, or neither. neither: $f(-x) = x^2 - x$, which is not equal to $f(x)$ or $-f(x)$

Summary (fill in): Substitute $-x$ into the function: if you get back the ORIGINAL function, it's even; if you get back the negative of the original, it's odd; otherwise, neither.